

Filobateer Ghaly

PhD Student

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Education

Ph.D. in Particle Physics	📅 2024 (current) 📍 University of Calgary
M.Sc. in Space Physics	📅 2022 - 2023 📍 University of Calgary
B.Sc. in Physics	📅 2017 - 2021 📍 University of Calgary



Languages

Arabic	C2*	●●●●●●
English	C2*	●●●●●●

Research Contributions

HAICU Trap Commissioning – 2025–(present), Ph.D. Thesis

- Constructed the solenoidal Bitter magnets.
- Tested the quadrupoles of the trap with running cooling water and current up to 1200 Amps.
- Researching and developing strategies for integrating the thermistors into the quadrupoles for monitoring the temperature of the magnets during operation.

HAICU Experiment Simulations – 2024–(present), Ph.D. Thesis

- Developed COMSOL Multiphysics simulations in conjunction with C++ code to model the magnetic field used in the HAICU experiment.
- Integrated the field maps into a C++ software package to simulate the trajectories of hydrogen in our apparatus.
- Used the simulations to time the trapping of hydrogen, optimizing for maximum number of atoms trapped.
- Extended the simulations to include magnetic deceleration of hydrogen.

Radio Communication Disruption Forecasting – 2023–2024, M.Sc. Thesis

- Performed an ordinal statistical analysis of over 4 years of radio and riometer data.
- By finding correlations between the two datasets, I established the riometer response as proxy for high-frequency radio communication disruption.
- Identified the core drivers for signal loss for a wide range of frequencies from 5 to 15 MHz.
- Findings currently under review for publication in the *Journal of Geophysical Research: Space Physics*.

Sizing Radio-disrupted Regions – 2022–2023, M.Sc. Thesis

- Wrote a ray tracing program to model the undisturbed propagation of radio waves.
- Used the modelled signal, together with the signal status observed by the receiver, to estimate the size of the region where radio waves are disrupted.
- Extended the methodology to a network of multiple radio transmitters and receivers across Canada.
- The technique was published in: *Journal of Geophysical Research: Space Physics*: Ghaly et al., 2024. All code, data analysis, visualizations and writing were done by me. Coauthors helped with editing the paper.

Collaboration and Impact

Natural Resources Canada (NRCan) Collaboration – 2020–2024 Research Assistant and M.Sc. Thesis

- Upon discovering NRCan's underutilized time-signal transmitters, I verified their detectability in our data.
- I then established that their signals were disrupted by space weather, and collaborated with NRCan to incorporate their network of radio transmitters into my analysis.
- The success of this collaboration led to the development of a space weather forecasting/alerting system for modeling where and when radio communications are disrupted.

Software and Technical Development

MITACS Globalink Award: CERN Control System Revamp – Summer 2025

- I was awarded a MITACS Globalink Research Award to spend the summer of 2025 at CERN, revamping the control system for the microwaves in the ALPHA experiment.
- Upgrading from a National Instruments PXI crate to a compact-RIO, I developed entirely new control software in LabVIEW.
- The new software uses the Remote Action Engine framework, allowing for remote control and monitoring of the microwave system.

Real-time Data Pipeline and Automatic Baselineing – 2021–2023, Research Assistant and M.Sc. Thesis

- Adapted the Perseus, Software-Defined Radio (SDR), making it usable for research.
- Wrote software for data acquisition in Python and Java and tested the instrument's ability to function as a receiver.
- Developed an online data browser for the prototype instruments deployed to generate curated, raw (K0-level) data for the public. Back-processed and published K0 data for 2022 and 2023.
- Wrote instrument-side code to generate curated K0 data on-site for all instruments deployed. Data are now being published in real-time.
- Developed a Graphical User Interface (GUI) for semi-automatic, instrument-specific baselineing.
- Developed a methodology for real-time, automatic baselineing to generate operational (K2-level) data.
- Designed a web application with a Postgre database that browses through different specification parameters for flagging space weather events.

Publications



MSc Thesis

- Ghaly, F. (Sept. 2024). “Quantifying Ionospheric Electron Density Enhancements and their Impact on High Frequency Radio Wave Propagation”. en. URL: <https://hdl.handle.net/1880/119764>.



Articles

- Ghaly, F., E. Spanswick, R. Gillies, T. Cameron, S. Skone, R. a. D. Fiori, et al. (2024). “Use of Terrestrial High Frequency Signals in Riometer Data to Explore the Size of D-Region Electron Density Enhancements”. en. In: *Journal of Geophysical Research: Space Physics* 129.5. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1029/2023JA032375>, e2023JA032375. ISSN: 2169-9402. DOI: 10.1029/2023JA032375. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1029/2023JA032375>.



Reports

- Ghaly, F. and E. Spanswick (Mar. 2022). *Identification and Investigation of High Frequency Radio Wave Transmissions During Quiet and Disturbed Periods*. Tech. rep. Natural Resources Canada.



Conferences

- Ghaly, F., E. Spanswick, T. Cameron, et al. (Dec. 2023). “A Riometer Proxy for HF Terrestrial Link Status”. *American Geophysical Union (AGU) 2023 Fall Meeting*. San Francisco, California.
- Ghaly, F., E. Spanswick, R. Gillies, T. Cameron, S. Skone, R. A. D. Fiori, and A. Weatherwax (Feb. 2023). “Using Multi-frequency Riometers for Monitoring HF Signal Propagation And Modelling D-region Absorption”. *Division of Atmospheric and Space Physics Conferences (DASP)*. Canadian Space Agency (CSA). John H. Chapman Space Centre, Longueuil, Quebec.
- Ghaly, F., E. Spanswick, R. Gillies, T. Cameron, S. Skone, R. A. D. Fiori, A. Weatherwax, et al. (Dec. 2022). “Use of Terrestrial HF Signals in Riometer Data to Investigate the Size of the Absorbing Region”. *American Geophysical Union (AGU) 2022 Fall Meeting*. Chicago, Illinois.
- Ghaly, F., E. Spanswick, and R. Gillies (Feb. 2021). “Investigating Riometer Capacity for Terrestrial HF Signal Monitoring”. *Division of Atmospheric and Space Physics Conferences (DASP)*. Online via Zoom.

Skills

Software/ Programming: • Python • Java • JavaScript • C++ • MATLAB • GNU Radio Companion • PostgreSQL • SQLite • Django • CSS • HTML • COMSOL Multiphysics

Hardware: • Software Defined Radios (SDR) • Antennae • Microcontrollers

Modelling: • AutoCAD Inventor

Teaching Experience

Teaching Assistant (TA): Department of Physics and Astronomy  University of Calgary

Lab TA	PHYS 211: Mechanics		Fall, 2024
Lead TA	PHYS 355: Electromagnetic Theory		Winter, 2024
Lab TA	PHYS 397: Applied Physics Laboratory		Fall, 2023
Lab TA	PHYS 229: Modern Physics		Winter, 2021
Lead TA	PHYS 259: Electricity and Magnetism		Summer, 2020
Lab TA	PHYS 259: Electricity and Magnetism		Winter, 2020

Learning Assistant (LA): Department of Computer Science  University of Calgary

Tutorial LA	DATA 202: Thinking with Data		Winter, 2021
Tutorial LA	CPSC 501: Advanced Programming Techniques		Fall, 2020